FALI: Mechanistic Decomposition of Conditional Attention-Head Importance

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Abstract

Component importance in transformer circuits can change after another component is perturbed, but detecting such conditional importance does not explain why the change occurred. We introduce FALI (Factorized Attention-Level Intervention), an output-grounded counterfactual framework that decomposes the cross-state change in a selected attention head's causal usefulness into target scores, rival scores, other attention scores, value content, and surrounding-network context. Standard Owen attribution allocates interactions while reconciling the fine score decomposition with a coarse score/value/context game. Controlled synthetic systems and a randomly initialized transformer validate factor recovery, endpoint identities, intervention isolation, and coarse–fine consistency. In GPT-2 Small on indirect object identification, a genuinely recruited head exhibits a positive target-score contribution and a negative rival-score contribution, while another head shows similar attention adaptation without positive recruitment. We then preregister transfer of this fine-score mechanism to Pythia-160M induction. Conditional recruitment appears again, but the target and rival contributions reverse sign. Thus, across the two studied settings, a common functional response does not imply a common fine-grained mechanism. FALI provides a mechanism-discrimination layer for selected conditionally important attention heads under explicit intervention semantics.

Keywords: mechanistic interpretability, causal intervention, transformer circuits, attention heads, conditional importance

1 Introduction

A component's causal importance in a transformer need not be stable under intervention. Removing one component can change the contribution of others: downstream layers can compensate for ablated computation, individual attention heads can change functional role depending on which other heads are available, and components that appear unimportant in the intact model can become important after a primary circuit is removed (McGrath et al. 2023; McDougall et al. 2023; Rushing and Nanda 2024; Nam et al. 2025; Gong et al. 2026). This conditionality is not merely a complication of terminology. Conditional co-ablation can recover backup components that first-order ablation scores miss, while causal-mediation analyses of activation patching show that a component's measured effect can contain substantial interactions with the state of the rest of the model (Gong et al. 2026; Vaidyanathan et al. 2026).

These results sharpen a question that localization alone does not answer. Suppose an attention head is known to become more or less useful after another circuit component is perturbed. *What changed in and around that head that accounts for the shift in its causal usefulness?* A changed ablation effect does not distinguish among several possibilities. The head may route attention differently, the information carried by its value vectors may change, or an otherwise similar head output may become more or less useful because the surrounding network has changed. Even within changed routing, a task-relevant attention edge can strengthen because its own score increases, because rival scores decrease, or because score changes elsewhere alter the head's computation. Existing work provides important pieces of this picture—including conditional importance, interaction-aware attribution, and causal separation of query–key routing from value-mediated computation—but these pieces target different estimands (Gong et al. 2026; Vaidyanathan et al. 2026; Wang et al. 2026).

We introduce **FALI (Factorized Attention-Level Intervention)**, a hypothesis-driven framework for resolving such conditional changes in a selected attention head. FALI begins with two model states: an intact state A and a perturbed state B. For a candidate head whose task-relevant target edge or edge set has already been specified, FALI factorizes the cross-state change into five intervention variables: target scores T, rival scores R, other structurally valid attention scores O, value content V, and surrounding-network context C. The first three factors jointly determine the candidate's attention pattern; V determines the content available to be transported; and C switches the surrounding computation between the intact and perturbed model states. For each of the 2^5 hybrid states, FALI defines the candidate's utility as its output-grounded contribution to the task metric: the metric with the hybrid candidate output injected minus the metric with that candidate zeroed in the same surrounding context. Thus the object being explained is not an attention probability, but the candidate head's change in causal usefulness.

Interactions among these factors make a purely sequential attribution arbitrary. We therefore use the standard Owen value with the prospectively fixed coalition structure $\{\text{SCORE} = \{T, R, O\}, \text{VALUE} = \{V\}, \text{CONTEXT} = \{C\}\}$ (Owen 1977). This gives a fine decomposition into $\phi_T, \phi_R, \phi_O, \phi_V, \phi_C$ while preserving a coarse score/value/context decomposition: $\phi_T + \phi_R + \phi_O$ equals the SCORE contribution in the quotient game. We make no claim of mathematical novelty for the Owen

value. The methodological object is the counterfactual intervention game and the architecture-aligned factorization to which it is applied.

FALI is deliberately downstream of circuit discovery. Path and activation patching localize components or paths that affect behavior (Goldowsky-Dill et al. 2023; Zhang and Nanda 2024; Heimersheim and Nanda 2024); conditional methods such as CoAx ask which components become important after a primary set is removed (Gong et al. 2026); interaction analyses establish that measured causal effects can depend on the rest of the network (Vaidyanathan et al. 2026). FALI instead takes a selected conditional change as its explanandum and asks which changes in the candidate's score computation, value content, or surrounding context account for it under an explicit intervention model. To our knowledge, FALI is the first framework to decompose the intervention-induced change in an attention head's output-grounded causal usefulness into architecture-aligned score, value, and surrounding-context contributions while resolving the score contribution into target, rival, and other-score channels (Gong et al. 2026; Vaidyanathan et al. 2026; Wang et al. 2026; Zhu et al. 2026). This claim is narrower than separating routing from value content itself, which is established in transformer-circuit analyses and causal attention decompositions (Elhage et al. 2021; Franco and Crovella 2025; Wang et al. 2026).

We validate the intervention semantics before using FALI on pretrained models. Deterministic synthetic attention systems isolate each of T, R, O, V, C in turn and test a known cross-factor interaction; a small randomly initialized transformer then checks score-to-pattern reconstruction, value-to-head-output reconstruction, endpoint identities, batch-specific score partitions, absence of live-candidate leakage, and agreement between the fine Owen decomposition and its coarse quotient game. These tests are intended to establish that the implemented decomposition identifies the factor manipulated in controlled systems, not that the chosen factorization is the unique natural decomposition of transformer computation.

We then study two substantively different settings. In GPT-2 Small on indirect object identification (IOI) (Wang et al. 2023), corrected functional analysis first shows why attention adaptation cannot be used as a proxy for recruitment: L10H2 becomes more causally useful after the canonical primary Name Movers are removed, whereas L10H10 increases target attention strongly while its total causal usefulness decreases. On broad canonical IOI templates, FALI assigns L10H2 a positive target-score contribution and a negative rival-score contribution; the same target-positive/rival-negative fine-score pattern also appears for L10H10 despite its negative total conditional gain. Tested matched-mean and matched-resample intervention variants preserve the preregistered fine-channel result, while a broader preregistered robustness gate fails because its zero-damage reference criterion is not satisfied; we report both outcomes.

Finally, we test a preregistered transfer of the GPT-2 fine-score mechanism to a different model family and task: Pythia-160M induction, using an established induction setting (Olsson et al. 2022). Conditional recruitment appears again, but the fine-score signs reverse: the target contribution is negative and the rival contribution positive. The transfer hypothesis therefore fails. Across the two studied settings, conditional recruitment is shared while the preregistered fine-score mechanism is not. This negative transfer result is central to the paper: it demonstrates the value of distinguishing

a common functional label from the mechanism that produces it, rather than treating recruitment itself as evidence for a single internal explanation.

Our contributions are threefold:

1. We define an output-grounded counterfactual intervention game for explaining the cross-state change in a selected attention head's causal usefulness.
2. We factor that change into score, value, and surrounding-context contributions, with target/rival/other-score resolution inside SCORE and coarse–fine consistency via standard Owen attribution.
3. We show that the same high-level conditional-recruitment phenomenon can conceal different fine mechanisms: GPT-2 IOI and Pythia induction exhibit opposite target/rival score signs under the specified FALI games.

2 Related Work

2.1 Self-repair and conditional component importance

Transformer circuits can adapt when components are ablated. The IOI circuit analysis of Wang et al. (2023) identified primary and backup Name Mover heads in GPT-2 Small, providing an early concrete setting in which several attention heads participate in a behavior and some become especially relevant under intervention. Subsequent work established self-repair more broadly. The Hydra effect describes compensatory changes in downstream language-model computation after ablation (McGrath et al. 2023). Copy-suppression analysis showed that self-repair can arise through specific functional mechanisms rather than a single generic backup process (McDougall et al. 2023). Rushing and Nanda (2024) further found self-repair across model families and the full training distribution, with substantial prompt-level variability and contributions from mechanisms including LayerNorm scaling and anti-erasure.

FALI does not claim the discovery of self-repair or heterogeneous compensation. More directly, it also does not introduce the idea of conditional component importance. Gong et al. (2026) formalize Conditional Co-Ablation (CoAx), which scores how much a remaining unit's ablation effect grows once a primary set has been removed. Their experiments show that this conditional score can recover backup heads missed by first-order rankings and transfers from GPT-2 IOI to induction settings across multiple models. This establishes both the empirical relevance and the practical utility of asking how component importance changes under intervention.

The distinction is therefore one of explanatory stage. Conditional ablation answers whether, or to what extent, a component becomes more important in a perturbed circuit. FALI assumes such a candidate has already been selected and asks which changes in the candidate's attention-score computation, value content, and surrounding context account for that cross-state change. In this sense CoAx and FALI are complementary: the former can identify *which* components warrant conditional analysis, whereas FALI provides a structured mechanism-resolution experiment for a chosen attention head.

2.2 Interaction-aware causal attribution

Activation patching and related interchange interventions are standard tools for testing whether internal activations causally contribute to model behavior. Path patching refines this idea to hypotheses about particular computational paths (Goldowsky-Dill et al. 2023); systematic studies have also shown that patching conclusions depend on choices such as corruption, metric, and intervention design (Zhang and Nanda 2024; Heimersheim and Nanda 2024). These methods are primarily localization or causal-testing tools: they identify components, activations, or paths whose replacement changes an outcome. FALI uses the same broad experimental tradition but addresses a different estimand—the change in one selected head's output-grounded causal contribution between two circuit states.

A particularly close conceptual neighbor is Vaidyanathan et al. (2026). Re-deriving activation patching through causal mediation analysis, they show that a component's natural indirect effect can include interaction terms that measure how its effect depends on the state of other model components. In GPT-2 IOI these interactions can hide conditionally important components, inflate others, and contribute to instability in component rankings. This result rules out any novelty claim that FALI is the first method to observe context-dependent causal effects. Instead, it motivates the next question: once a conditional or interaction-dependent effect has been identified, can we resolve that change into architecture-level factors of the selected attention head?

Causal Head Gating provides a complementary head-level perspective. Nam et al. (2025) learn gates over attention heads, classify their roles as facilitating, interfering, or irrelevant, and find that head roles can depend on interactions with other heads. FALI is neither a learned gating method nor a global head-role taxonomy. It fixes one candidate head and two model states, then intervenes on explicitly defined internal factors of that head and the surrounding model. Likewise, we do not claim that FALI is generally more faithful than patching: intervention-based interpretability results can be sensitive to methodology, as demonstrated by circuit-faithfulness studies (Miller et al. 2024). Our robustness experiments therefore support only the intervention variants actually tested.

These interaction-aware results also inform our interpretation of the context factor C. Under FALI, C is not a localized "downstream sensitivity" variable. It denotes the surrounding model computation that differs between intact and perturbed contexts outside the explicitly injected candidate score/value state. It can therefore aggregate changes in other heads, MLPs, residual representations, normalization, and upstream or downstream computation. A large ϕ_C is evidence about the specified counterfactual game, not identification of a unique external mechanism.

2.3 Decomposing attention computation

The separation between attention routing and transported content is foundational rather than novel to FALI. The transformer-circuits framework distinguishes the query–key (QK) circuit, which determines attention scores and patterns, from the output/value (OV) circuit, which determines what information is written when a source position is attended to (Elhage et al. 2021). Later causal work makes increasingly

fine distinctions inside attention computation. Franco and Crovella (2025) identify low-dimensional communication signals with a causal relationship to attention patterns, while Wang et al. (2026) causally separate Query–Key routing from Value updates when explaining how few-shot examples shape function vectors. Recent factorial intervention work likewise shows that attention allocation and token content can be experimentally separated yet strongly coupled in a different robustness setting (Zhu et al. 2026).

FALI builds on these distinctions but changes the object being decomposed. Its dependent quantity is not the attention pattern itself, nor the quality of a function vector, but the *cross-state change in the selected head's downstream causal usefulness*. SCORE and VALUE are therefore accompanied by an explicit CONTEXT factor. Within SCORE, FALI further partitions structurally valid score coordinates into task-target scores T, rival scores R in the same task-relevant query row, and all other valid scores O. This target/rival/other partition is hypothesis-driven: it requires a task-relevant target edge or edge set to be specified in advance, and is not an automatic circuit-discovery procedure.

Because score, value, and context can interact non-additively, FALI uses the standard Owen value for a game with a priori coalition structure (Owen 1977). The coalition structure is fixed as SCORE=$\{T, R, O\}$, VALUE=$\{V\}$, and CONTEXT=$\{C\}$. Owen attribution is established game-theoretic machinery; our use of it is motivated by the requirement that the fine score contributions reconcile with the coarse SCORE contribution rather than by a claim of a new attribution theorem. The resulting values should also be read relative to the intervention semantics. Hybrid internal states need not be states the unmodified model would naturally visit, and causal-intervention research has documented that representation-level interventions can move models away from their natural internal distributions (Grant et al. 2025). We therefore treat FALI as a controlled counterfactual decomposition, not as evidence that the model implements five independent natural pathways.

Finally, our emphasis on explicit validation reflects a broader evaluation problem in mechanistic interpretability. Benchmarks such as MIB prioritize precise recovery of causal pathways or variables rather than plausible narrative alone (Mueller et al. 2025). FALI's synthetic single-factor systems and implementation invariants serve the narrower purpose of checking that the intervention pipeline recovers known manipulated factors before empirical use. They do not remove the substantive limitations of target selection, coalition choice, or intervention dependence, which we address explicitly in Section 9.

3 Problem Formulation

3.1 Conditional causal usefulness

Let f be a transformer evaluated by a scalar task metric m. We consider two executions of the same task instance: an *intact* state A and a *perturbed* state B, obtained by applying a prespecified intervention to a primary circuit component or set of components. Let h denote a candidate attention head whose change in functional importance we want to explain. The task metric can differ across applications—for

example, an IO-minus-subject logit difference in IOI or the target-token log probability in induction—but it is fixed within an analysis.

For context $s \in \{A, B\}$, let m_s^h be the task metric with the candidate head operating normally and let m_s^0 be the metric when that candidate is zeroed while all other aspects of context s are held fixed. We define the candidate's output-grounded causal usefulness in context s as

$$E_s(h) = m_s^h - m_s^0. \tag{1}$$

The intact and conditional effects are therefore $E_A(h)$ and $E_B(h)$, respectively. The cross-state change we seek to explain is

$$\Delta_h = E_B(h) - E_A(h). \tag{2}$$

In our experiments this quantity is called *conditional gain*. Positive Δ_h means that the candidate's contribution becomes more positive after the primary intervention; it does not by itself imply that the candidate is supportive in the perturbed state. We reserve *positive functional recruitment* for cases in which both $E_B(h) > 0$ and $\Delta_h > 0$. FALI itself is not restricted to positive recruitment: it decomposes positive or negative changes in causal usefulness.

This definition deliberately evaluates the candidate against a zero-candidate baseline in the *same* surrounding context. Consequently, a difference between $E_A(h)$ and $E_B(h)$ cannot be produced merely by a shift in the task metric that affects both the candidate-present and candidate-zero conditions equally. The estimand is the change in the candidate's marginal causal contribution under the specified intervention, not the raw performance difference between states A and B.

3.2 Mechanism-resolution objective

The problem begins after a candidate head and a task-relevant attention target have been chosen. They may come from prior circuit analysis, conditional co-ablation, patching, or a task-specific hypothesis. FALI does not search over heads, discover a target edge, or infer a circuit automatically. Its input is a selected head h, two model states A and B, a scalar task metric m, and a target query–key edge (or prespecified target edge set) whose score is meaningful for the task.

The objective is to account for Δ_h using changes in three architecture-level objects associated with the candidate: its pre-softmax attention scores, its value vectors, and the surrounding computation in which the resulting head output is used. The score tensor is then resolved more finely into the task-target score, rivals in the same query row, and scores at other query positions. This yields a controlled mechanism-discrimination question: under a fixed intervention algebra, which candidate-local or surrounding changes receive credit for the observed change in output-grounded usefulness? Figure 1 summarizes this distinction.

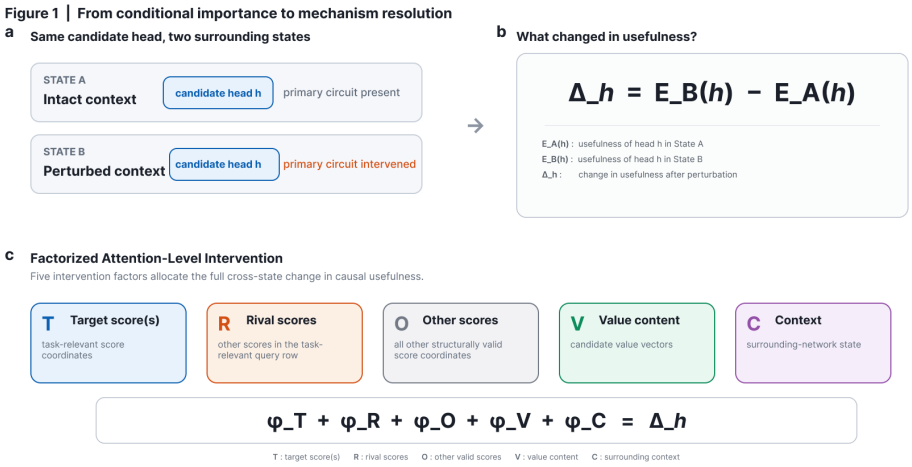


Fig. 1 FALI resolves a conditional change in a selected attention head's causal usefulness. (a) The same candidate head h is evaluated in an intact state A and a prespecified perturbed state B. (b) The quantity to be explained is the cross-state change $\Delta_h = E_B(h) - E_A(h)$ in output-grounded causal usefulness. (c) FALI attributes this endpoint change under an explicit intervention game across task-target scores (T), rival scores (R), other structurally valid scores (O), value content (V), and surrounding-network context (C), with Owen efficiency $\phi_T + \phi_R + \phi_O + \phi_V + \phi_C = \Delta_h$. The candidate head and task-relevant target coordinates are specified before FALI analysis.

4 Factorized Attention-Level Intervention

4.1 Architecture-aligned factors

For one candidate head and one task instance, let S^A and S^B denote its cached pre-softmax attention-score tensors in the intact and perturbed executions, and let V^A and V^B denote its corresponding value tensors. Let $q^\star$ be the task-relevant query position and $k^\star$ the target key position. We use an explicit structural validity mask

$$\mathcal{M} = \{(q, k) : q < \ell,\ k \le q\}, \tag{3}$$

for a right-padded causal sequence of valid length ℓ; padding and future positions are structural constants rather than intervention variables. The valid score coordinates are partitioned into

$$T = \{(q^\star, k^\star)\}, \tag{4}$$
$$R = \{(q^\star, k) \in \mathcal{M} : k \neq k^\star\}, \tag{5}$$
$$O = \mathcal{M} \setminus (T \cup R). \tag{6}$$

Thus T, R, and O are pairwise disjoint and cover every structurally valid score coordinate exactly once. When an application specifies multiple target edges, T can be generalized to that prespecified set and R defined as the remaining valid coordinates

in the corresponding task-relevant query row(s); the experiments in this paper use one target coordinate per example.

FALI has five binary factors, each taking its value from state A (coded 0) or state B (coded 1):

- T: task-target attention score(s);
- R: rival scores in the same task-relevant query row;
- O: all other structurally valid scores of the candidate head;
- V: candidate value vectors at all valid source positions;
- C: surrounding-network context, intact versus perturbed.

The first three factors jointly determine the complete candidate attention pattern. V controls the content combined by that pattern. C controls the rest of the model execution outside the explicitly injected candidate state. In particular, C is not defined as a uniquely localized downstream mechanism; it can aggregate differences in other attention heads, MLPs, residual representations, normalization, and upstream or downstream computation.

4.2 Counterfactual intervention game

Let $(\tau, \rho, \omega, \nu, c) \in \{0,1\}^5$ denote one FALI cell. For a binary state index b, write $S^b = S^A$ if $b = 0$ and $S^b = S^B$ if $b = 1$, with the analogous convention for V^b. We construct a hybrid score tensor by taking each coordinate block from its selected endpoint,

$$S_{\tau\rho\omega} = \mathbf{1}_T \odot S^\tau + \mathbf{1}_R \odot S^\rho + \mathbf{1}_O \odot S^\omega, \tag{7}$$

with invalid coordinates fixed to the structural attention mask. We then recompute the attention pattern from the hybrid scores,

$$P_{\tau\rho\omega} = \text{masked_softmax}(S_{\tau\rho\omega}), \tag{8}$$

and reconstruct the candidate head output using the selected value tensor,

$$z_{\tau\rho\omega\nu}[q] = \sum_k P_{\tau\rho\omega}[q,k] V^\nu[k]. \tag{9}$$

The reconstructed z is injected for the candidate head at all valid query positions. This full-output overwrite is important: the live candidate computation is not allowed to leak into a hybrid cell.

Let $F_c[z_h \leftarrow z]$ denote a forward pass in surrounding context c, with $c = 0$ selecting state A and $c = 1$ selecting state B, and with the full output of candidate head h replaced by z. The FALI utility for a cell is

$$U(\tau, \rho, \omega, \nu, c) = m(F_c[z_h \leftarrow z_{\tau\rho\omega\nu}]) - m(F_c[z_h \leftarrow 0]). \tag{10}$$

The zero-candidate comparison is always evaluated in the same context c as the hybrid injection. FALI therefore contains $2^5 = 32$ counterfactual cells.

The two complete endpoints recover the effects in Eq. 1:

$$U(0,0,0,0,0) = E_A(h), \tag{11}$$
$$U(1,1,1,1,1) = E_B(h), \tag{12}$$

so that

$$\Delta_h = U(1,1,1,1,1) - U(0,0,0,0,0). \tag{13}$$

This endpoint identity is a required implementation invariant, not an approximation in the mathematical definition.

Most mixed cells need not correspond to an activation state encountered during an unmodified forward pass. FALI should therefore be read as an explicit counterfactual/interchange-intervention game. Its attributions are conditional on the chosen endpoint states, target partition, masking semantics, and intervention construction; they are not claims that the model naturally executes five separable causal pathways. Figure 2 depicts the 32-cell construction.

4.3 Standard Owen attribution and coarse–fine reconciliation

Let the player set be $N = \{T, R, O, V, C\}$ and associate each coalition $K \subseteq N$ with the cell in which factors in K take their B state and all remaining factors take their A state. This defines the characteristic function

$$v(K) = U(\mathbf{1}_K). \tag{14}$$

Because the score factors are subdivisions of one computational object, whereas value and context are separate coarse factors, we fix the coalition structure

$$\Pi = \{\text{SCORE} = \{T, R, O\},\ \text{VALUE} = \{V\},\ \text{CONTEXT} = \{C\}\}. \tag{15}$$

The partition membership is fixed; the block order is not. We use the standard Owen value, which averages marginal contributions over all permutations in which members of each block remain contiguous (Owen 1977). For this partition there are $3! \times 3! = 36$ admissible permutations.

For completeness, let $G \in \Pi$ be the block containing player i, let $M = |\Pi| = 3$, let $\mathcal{B} \subseteq \Pi \setminus \{G\}$ be a subset of predecessor blocks, and let $A_i \subseteq G \setminus \{i\}$ be a subset of predecessor players within G. Define $W_{\mathcal{B}} = \bigcup_{H \in \mathcal{B}} H$. The Owen value is

$$\phi_i = \sum_{\mathcal{B} \subseteq \Pi \setminus \{G\}} \frac{|\mathcal{B}|!\,(M - |\mathcal{B}| - 1)!}{M!} \sum_{A_i \subseteq G \setminus \{i\}} \frac{|A_i|!\,(|G| - |A_i| - 1)!}{|G|!} \tag{16}$$
$$\times \left[v(W_{\mathcal{B}} \cup A_i \cup \{i\}) - v(W_{\mathcal{B}} \cup A_i) \right]. \tag{17}$$

This is equivalent to averaging sequential marginal contributions over the 36 admissible permutations. We use independent subset-weight and permutation implementations in validation.

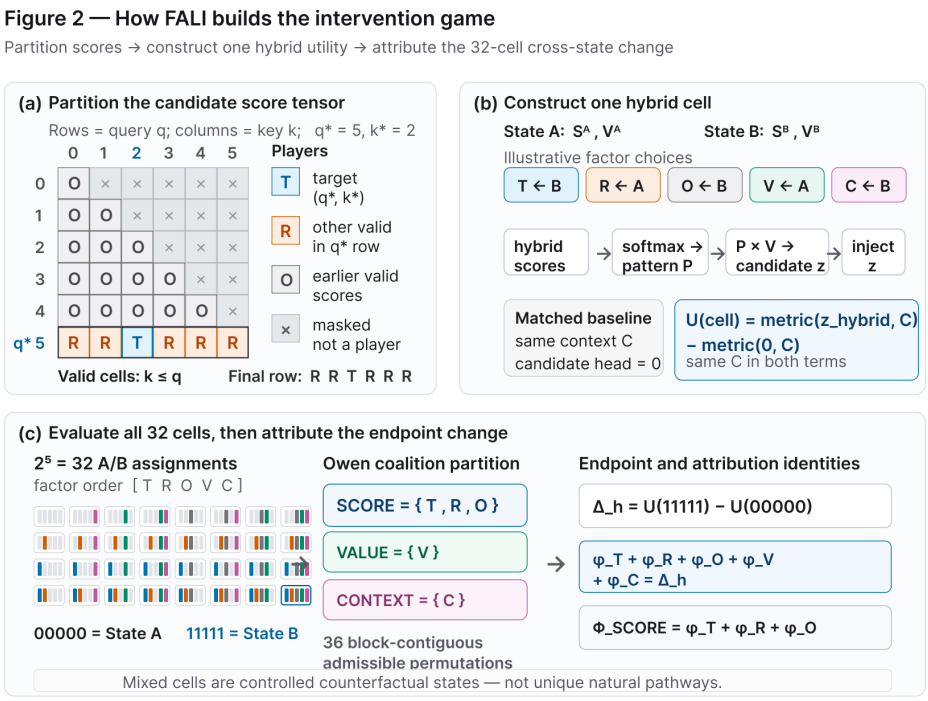


Fig. 2 FALI constructs a 32-cell counterfactual intervention game for the selected attention head. (a) Structurally valid pre-softmax scores are partitioned into the prespecified task-target coordinate(s) T, rival scores R in the same task-relevant query row, and all other valid scores O; masked coordinates are not players. (b) Each hybrid cell selects T, R, O, V, and C from the intact or perturbed endpoint, reconstructs the candidate attention pattern and head output, and defines utility relative to a candidate-zero run in the same surrounding context C. (c) All $2^5 = 32$ hybrid states are evaluated. Standard Owen attribution under SCORE $= \{T, R, O\}$, VALUE $= \{V\}$, and CONTEXT $= \{C\}$ allocates the endpoint change over 36 block-contiguous admissible permutations, satisfying $\sum_i \phi_i = \Delta_h$ and $\Phi_{\text{SCORE}} = \phi_T + \phi_R + \phi_O$. The mixed cells are controlled counterfactual states, not claims that the model naturally implements five independent causal pathways.

The first property needed for interpretation is efficiency:

$$\phi_T + \phi_R + \phi_O + \phi_V + \phi_C = v(N) - v(\varnothing) = \Delta_h. \tag{18}$$

The attributions therefore allocate the entire endpoint change in causal usefulness, including interaction effects, under the specified game. They should not be interpreted as pure direct effects.

The second property gives the coarse–fine bridge. Define the quotient game on the three meta-players by

$$q(\mathcal{K}) = v\left(\bigcup_{G \in \mathcal{K}} G\right), \qquad \mathcal{K} \subseteq \Pi. \tag{19}$$

Standard Owen group consistency gives

$$\Phi_{\text{SCORE}} = \phi_T + \phi_R + \phi_O = \text{Shapley}_q(\text{SCORE}), \tag{20}$$

$$\phi_V = \text{Shapley}_q(\text{VALUE}), \tag{21}$$

$$\phi_C = \text{Shapley}_q(\text{CONTEXT}). \tag{22}$$

When the whole SCORE block switches together, its hybrid scores reproduce the complete A or B candidate pattern. The quotient SCORE/V/C game therefore matches the earlier pattern/value/context intervention game at the corresponding cells; this numerical bridge is tested explicitly in Section 5.2. The purpose of Owen attribution here is not mathematical novelty but consistency between a coarse score/value/context explanation and the finer $T/R/O$ explanation inside SCORE.

Finally, the historical two-factor attention-probability version of FALI is nested as a special case rather than identified with the present causal attributions. If the utility is changed from output-grounded usefulness to target attention probability, O, V, and C are held fixed, and only T and R vary, Eq. 17 reduces to the ordinary two-player Shapley decomposition of target-score versus competing-score changes. The present method changes both the intervention scope and, crucially, the quantity being explained.

5 Method Validation

Before applying FALI to pretrained models, we tested whether the intervention game recovers known factors in controlled systems and reproduces ordinary transformer computation at its two natural endpoints. These preregistered checks validate implementation semantics rather than learned-model mechanisms. Figure 3 summarizes the tests.

5.1 Analytical and synthetic ground truth

We constructed deterministic attention systems containing causal scores, masked softmax, values, $z = PV$, and a context-specific readout, and evaluated all 32 FALI cells. In five single-channel cases, only the manipulated factor received attribution: for example, the target-only system gave $\Delta_h = 0.268952$ with $\phi_T = 0.268952$, and the rival-only system gave $\Delta_h = -0.494100$ with only ϕ_R nonzero. The O, V, and C cases behaved analogously; the largest attribution error was 1.11×10^{-16}. Panel (a) of Fig. 3 shows this factor-isolation recovery directly.

A separate pure $T \times C$ construction had $\Delta_h \approx 2$ with no isolated marginal effect and yielded the analytically expected Owen split $\phi_T \approx 1$, $\phi_C \approx 1$, with $R, O, V = 0$. All-A and all-B hybrids reconstructed their endpoints exactly at recorded precision, and collapsing T, R, O into SCORE reproduced the corresponding three-player SCORE/VALUE/CONTEXT decomposition with maximum discrepancy 1.11×10^{-16}. Panel (b) of Fig. 3 shows the expected interaction allocation.

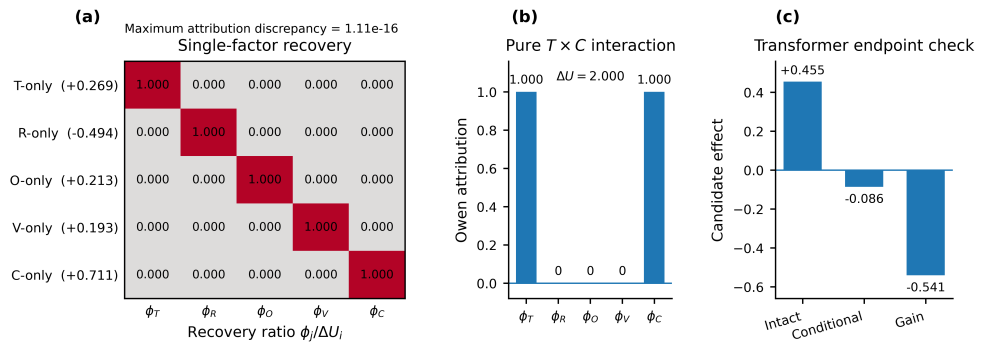


Fig. 3 Controlled validation recovers known factors, interaction allocation, and transformer endpoint identities. (a) In five deterministic single-factor systems, the manipulated factor receives the full endpoint change: each row shows the recovery ratio $\phi_j/\Delta U_i$, with the raw injected ΔU_i given beside the row. The maximum attribution discrepancy is 1.11×10^{-16}. (b) A pure $T \times C$ construction with $\Delta U \approx 2$ yields the analytically expected Owen split $\phi_T \approx 1$ and $\phi_C \approx 1$, with $R, O, V = 0$. (c) In the randomly initialized two-layer TransformerLens validation model, independently measured intact effect, conditional effect, and conditional gain are 0.455046, -0.085684, and -0.540729, respectively, and the corresponding FALI endpoint cells reproduce them exactly at recorded precision. Reconstruction, leakage, and model-level quotient residuals are zero at recorded precision. These are implementation and controlled-ground-truth checks, not evidence that mixed FALI hybrids are unique natural model pathways.

5.2 Transformer implementation invariants

We then repeated the mechanical checks in a randomly initialized two-layer TransformerLens model (two heads per layer, $d_{\text{model}} = 32$, $d_{\text{head}} = 16$, $d_{\text{mlp}} = 64$, context length 8, vocabulary 48), evaluated in FP32 on CPU with seed 4242. Masked softmax reconstructed cached intact and perturbed attention patterns exactly at recorded precision, pattern–value multiplication reconstructed candidate z, masked coordinates received zero probability, and batch-specific $T/R/O$ partitions had one target coordinate, zero overlap, and complete valid-score coverage per example.

Independent ordinary forward passes gave intact effect 0.455046, conditional effect -0.085684, and conditional gain -0.540729; the corresponding FALI endpoint cells reproduced all three with zero recorded discrepancy. Full candidate-z injection left non-candidate heads unchanged, and large perturbations to the candidate's live query computation, value computation, or both produced zero recorded leakage once z was overwritten. The model-level fine game and collapsed SCORE/VALUE/CONTEXT quotient also agreed exactly. Panel (c) of Fig. 3 reports the independent endpoint values; reconstruction, leakage, and model-level quotient discrepancies are zero at recorded precision.

These checks establish factor isolation, structural masking, endpoint recovery, intervention isolation, and coarse–fine reconciliation for the implemented pipeline. They do not establish that the factorization is unique, that mixed hybrids are on-manifold, or that a large attribution identifies a unique natural causal pathway; those interpretation limits remain explicit in Section 9.

6 GPT-2 IOI: Conditional Recruitment and Fine-Score Decomposition

6.1 Experimental design

We apply FALI to GPT-2 Small on indirect object identification (IOI), using the canonical primary Name Movers L9H6, L9H9, and L10H0 as the perturbed set and final-position IO-minus-subject logit difference as the task metric (Wang et al. 2023). The confirmatory cohort contains 15 low-level templates crossed with BABA/ABBA structure (30 strata), 30 prompts per stratum, and four independent seeds (42, 1, 8, 22), for 900 prompts per seed and 3600 total. L10H2 and L10H10 are the two primary candidates; L9H7 and L9H0 are same-layer architectural controls.

Confirmation is specific to the previously unseen fine channels ϕ_T, ϕ_R, and ϕ_O. The preregistered directional hypotheses were $\phi_T > 0$ and $\phi_R < 0$ for each target head, with ϕ_O descriptive only. Point estimates equally weight strata within seed and then seeds; 95% intervals use 10,000 deterministic hierarchical template-cluster bootstrap draws with shared template resampling across seeds and prompt resampling within seed–template–pattern cells. A preregistered sensitivity removes template 0, which overlaps a legacy discovery template, leaving 14 clusters and 28 strata.

6.2 Conditional recruitment is not attention increase

L10H2 is positively recruited on the broad canonical cohort: its conditional effect is 0.382 (95% CI $[0.341, 0.424]$) and its conditional gain is 0.298 ($[0.263, 0.335]$). Target attention alone does not identify this response. L10H2's target-attention increase is 0.279 ($[0.262, 0.296]$), whereas L10H10 increases target attention even more, by 0.310 ($[0.285, 0.335]$), despite a negative conditional gain of -0.263 ($[-0.302, -0.225]$) and negative conditional effect of -0.079 ($[-0.111, -0.046]$). Panel (a) of Fig. 4 makes this dissociation explicit: both candidates increase target attention, while their conditional gains lie on opposite sides of zero.

The earlier corrected two-template screen had found L10H10 still supportive but downregulated; on the broader canonical distribution its mean conditional effect crosses below zero. The stable conclusion is therefore distributionally bounded but clear: a strong target-attention increase is not sufficient evidence of positive functional recruitment. FALI consequently explains the output-grounded conditional change rather than the attention change itself.

6.3 Canonical FALI mechanism

For L10H2, $\Phi_{\mathrm{SCORE}} = 0.445$, $\phi_V = -0.00035$, and $\phi_C = -0.146$. Within SCORE, the target contribution is strongly positive, $\phi_T = 0.783$ (95% CI $[0.722, 0.838]$), while the rival contribution is negative, $\phi_R = -0.341$ ($[-0.372, -0.311]$); $\phi_O = 0.0039$. All four seed means and all 30 stratum means have the preregistered $T+/R-$ signs.

L10H10 shows the same fine-score signs despite its negative total gain: $\phi_T = 0.329$ ($[0.300, 0.355]$), $\phi_R = -0.126$ ($[-0.139, -0.113]$), and $\phi_O = 0.0036$. Its positive SCORE contribution (0.206) is outweighed by a negative surrounding-context contribution ($\phi_C = -0.471$), with near-zero value contribution. Thus a favorable-looking

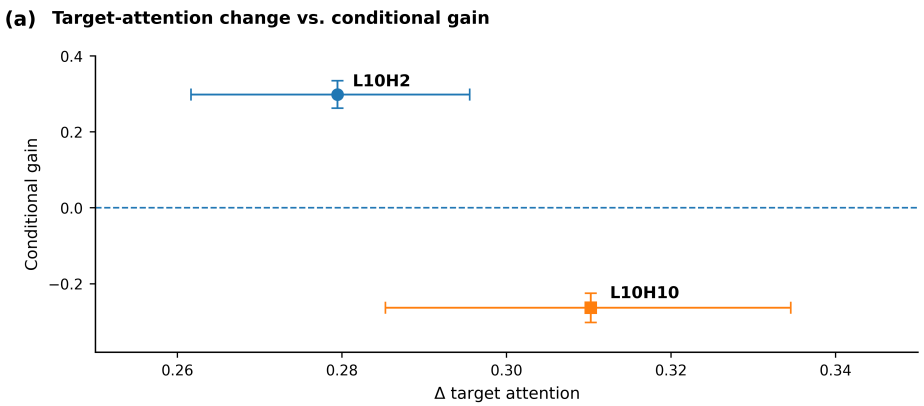


Fig. 4 Target-attention adaptation is not sufficient for positive functional recruitment in GPT-2 IOI. (a) L10H2 and L10H10 both increase attention to the indirect-object target after ablation of the canonical primary Name Movers, but their conditional gains have opposite signs: L10H2 becomes more causally useful whereas L10H10 becomes less useful on the broad canonical distribution. Points are exact balanced means; horizontal and vertical intervals are hierarchical 95% bootstrap confidence intervals over the canonical template design.

score adaptation under the specified game need not imply functional recruitment. Panel (b) of Fig. 4 shows the fine-factor profiles, while panel (c) shows the cumulative SCORE/VALUE/CONTEXT decomposition that drives L10H10's net gain below zero.

The preregistered EF-4 fine-score gate passes for both heads. All four directional hypotheses hold in every seed and all 30 strata, and the template-0-excluded sensitivity retains the predicted directions with CIs excluding zero for both heads. The same-layer controls L9H7 and L9H0 have zero recorded $T/R/O/V$ attributions, with their endpoint changes assigned to C up to numerical residual 1.11×10^{-16}. These results support a stable GPT-2 mechanism pattern under the tested game, not a universal recruitment mechanism: L10H10 already shows that the same fine-score signs can occur without positive recruitment, and Owen values allocate interactions rather than pure direct effects.

6.4 Intervention robustness

EF-5 tested matched-mean and matched-resample alternatives to the canonical zero ablation. The preregistered fine channels retained their directions under both schemes, with hierarchical CIs excluding zero and the predicted sign in all 30 strata. Under matched mean, L10H2 had $\phi_T = 0.440$, $\phi_R = -0.166$ and L10H10 $\phi_T = 0.188$, $\phi_R = -0.0816$; under matched resample the corresponding values were $0.265, -0.0474$ and $0.0858, -0.0150$.

The overall EF-5 gate nevertheless *failed*. Its zero-ablation reference criterion required a positive damage-drop CI, but the observed reference was -0.319 with CI $[-0.396, -0.244]$, triggering the preregistered stop condition. We therefore claim robustness only for the fine-channel directions under the tested matched alternatives, not robustness to arbitrary ablations or an overall EF-5 pass.

7 Cross-Setting Test: Pythia Induction

7.1 Preregistered transfer design

We preregistered one transfer test of the GPT-2 target-positive/rival-negative pattern using Pythia-160M on repeated-token induction. For offsets $j \in \{8, 16, 24\}$, the input is $[x_0, \ldots, x_{31}, x_0, \ldots, x_j]$, the target is x_{j+1} at final query $q = 32 + j$, and the FALI target key is $k = j + 1$; a matched control replaces the first block with disjoint tokens. The transfer hypothesis predicted $\phi_T > 0$ and $\phi_R < 0$ for a later head that became more useful after an earlier induction head was removed. Failure of either sign counted against transfer.

Head selection was separated from FALI attribution. Discovery used seed 42 only (128 examples per offset; 384 total) and was forbidden from inspecting FALI attributions. Eligibility required a causally useful primary head and a later candidate with positive conditional effect and gain; the frozen selection rule chose L5H0 as primary and L8H2 as candidate. Confirmation then used seeds 1, 8, and 22, again with 128 examples per offset (1152 total). Estimates balance offsets within seed and seeds equally; 95% intervals use 10,000 deterministic bootstrap draws within seed–offset cells. The five frozen gates were induction behavior (G0), primary causal importance (G1), positive conditional recruitment (G2), transferred $\phi_T > 0$ (G3), and transferred $\phi_R < 0$ (G4).

7.2 Positive recruitment recurs while the fine mechanism reverses

The held-out cohorts satisfy the behavioral and functional prerequisites. Induction advantage is 14.017 (95% CI $[12.280, 15.939]$), the L5H0 primary drop is 1.146 ($[0.887, 1.539]$), and L8H2 is positively recruited: conditional effect 0.621 ($[0.456, 0.860]$) and conditional gain 0.486 ($[0.326, 0.722]$). These quantities have the required direction in every confirmation seed and offset, so G0–G2 confirm independently of the fine-score hypotheses. Panel (a) of Fig. 5 shows the shared functional result: conditional gain is positive in both the GPT-2 IOI and Pythia induction cases.

The fine mechanism reverses. FALI gives $\phi_T = -0.144$ (95% CI $[-0.247, -0.084]$) and $\phi_R = +0.118$ ($[0.058, 0.219]$), with the reversed sign in every seed and offset. This is not a near-zero failure: relative to the preregistered absolute score-channel denominator 0.261807, $|\phi_T|$ and $|\phi_R|$ account for approximately 54.9% and 45.0%, respectively. The remaining channels are small for O ($\phi_O = -0.00027$, descriptive) and V ($\phi_V = -0.00210$), while the surrounding-context contribution is positive and large ($\phi_C = 0.514$); as elsewhere, C is not interpreted as a uniquely localized external mechanism. Panel (b) of Fig. 5 contrasts the target, rival, and surrounding-context attributions directly with GPT-2 IOI and shows the sign reversal with preregistered confidence intervals.

Accordingly, EF-6 is FAIL: G0–G2 confirm, while G3 and G4 do not. No alternate pair, model, task, or threshold was substituted after this outcome. GPT-2 IOI and Pythia induction therefore both contain positive conditional recruitment in the studied cases, but their fine target/rival organizations are opposite. This falsifies transfer of

the preregistered GPT-2 fine-score mechanism to the tested Pythia setting; it does not establish that no shared recruitment mechanism exists more generally. Figure 5 therefore separates the shared functional recruitment result from the failed preregistered fine-mechanism transfer.

8 Discussion

8.1 A common functional response need not imply a common mechanism

The central empirical result is a failed mechanism-transfer hypothesis. GPT-2 IOI L10H2 is positively recruited after removal of the canonical primary Name Movers and has target-positive/rival-negative FALI score contributions; Pythia-160M induction L8H2 is also positively recruited after its primary intervention, but the preregistered fine-score signs reverse. Conditional recruitment therefore occurs in both studied settings while the specific GPT-2 fine-score organization does not transfer.

This distinction matters because *recruitment* names a change in causal usefulness, not the route by which it arises. Conditional-ablation methods can identify components whose importance changes after another component is removed (Gong et al. 2026), and interaction analyses show that component effects can depend strongly on the rest of the model (Vaidyanathan et al. 2026). FALI adds a mechanism-resolution step: under one explicit intervention game, two cases meeting the same functional criterion can receive opposite fine-grained score attributions.

GPT-2 also separates visible attention adaptation from recruitment. L10H10 increases target attention and has the same target-positive/rival-negative score pattern as L10H2, yet its broad-canonical conditional gain is negative. Thus neither attention change nor the sign of an individual FALI channel defines recruitment; the output-grounded candidate effect defines the functional response, and FALI allocates its cross-state change. Our evidence is limited to one GPT-2 IOI and one Pythia induction setting, so the reversal does not imply that fine mechanisms generally differ across models or that no shared mechanisms exist elsewhere.

8.2 Relationship to conditional attribution and patching

FALI is downstream of localization and conditional-importance analysis. Activation/path patching localizes behaviorally relevant activations or paths (Goldowsky-Dill et al. 2023; Zhang and Nanda 2024; Heimersheim and Nanda 2024); CoAx identifies components whose importance changes after perturbing a primary set (Gong et al. 2026); interaction-aware mediation analysis characterizes dependence on the surrounding network (Vaidyanathan et al. 2026). FALI instead assumes a selected attention head and target edge, then asks which changes in its score computation, value content, and surrounding context account for the change in output-grounded usefulness.

The method likewise does not claim novelty for separating routing from transported content or for Owen attribution. QK/OV structure and recent causal routing/value decompositions are established (Elhage et al. 2021; Franco and Crovella 2025; Wang et al. 2026); FALI applies architecture-aligned interventions to a different estimand,

the cross-state change in one head's downstream causal contribution, while explicitly including surrounding context. Standard Owen values provide internally consistent interaction allocation under the frozen coalition structure, not an intervention-invariant or unique natural causal decomposition. A practical workflow is therefore: identify a conditionally important component, then use FALI when the scientific question is why that selected attention head's usefulness changed.

9 Limitations

Hypothesis-driven scope.

FALI requires a selected attention head and task-relevant target edge or edge set. IOI and repeated-token induction provide clear target coordinates; less structured tasks may require substantial prior circuit analysis. The five-factor game evaluates all 32 cells per example and candidate, so it is intended for hypothesis-driven analysis of selected heads rather than exhaustive whole-model scanning. The method is also vulnerable to researcher degrees of freedom if candidates or target edges are chosen after inspecting FALI results; our confirmatory analyses mitigate this risk by freezing the relevant selections and directional hypotheses before confirmation. FALI is therefore mechanism resolution, not automatic circuit discovery.

Counterfactual hybrids.

Mixed $T/R/O/V/C$ cells need not occur in natural model execution. They answer what the model does under the specified interchange intervention, not which latent state it would otherwise visit. Representation-level interventions can move computation away from natural internal distributions (Grant et al. 2025); accordingly, FALI attributions are not evidence for five independent natural pathways.

Intervention and coalition dependence.

Attributions depend on the intact/perturbed endpoints, candidate-zero baseline, score partition, and SCORE/VALUE/CONTEXT coalition structure. Owen attribution allocates interactions consistently within that game, but another defensible intervention or grouping can allocate credit differently. In GPT-2, the target-positive/rival-negative channels persisted under the tested matched-mean and matched-resample alternatives, while the overall EF-5 gate failed its zero-damage reference criterion; we therefore make no claim of robustness to arbitrary ablations.

Coarse context factor.

C aggregates surrounding computation outside the injected candidate score/value state, potentially including other heads, MLPs, residual representations, normalization, and upstream or downstream changes. A large ϕ_C therefore assigns credit to this block of the game but does not identify the external component or pathway responsible. Decomposing C would require additional hypotheses and interventions.

Empirical and architectural scope.

The substantive evidence covers GPT-2 Small IOI and Pythia-160M repeated-token induction. This is sufficient to falsify transfer of the preregistered GPT-2 target/rival mechanism to the tested Pythia setting, but not to characterize recruitment mechanisms across architectures, scales, tasks, or training stages. The present algebra is also attention-head-specific; MLPs, residual features, or distributed mechanisms would require different architecture-aligned factors rather than mechanically reusing $T/R/O/V/C$.

10 Conclusion

FALI is an output-grounded counterfactual framework for explaining how a selected attention head's causal usefulness changes after a circuit perturbation. It separates target, rival, and other score changes from value content and surrounding-network context, with standard Owen attribution reconciling fine and coarse accounting.

The experiments show why this resolution matters. Attention adaptation need not imply positive recruitment, and positive recruitment need not share one fine mechanism: GPT-2 IOI and Pythia induction both exhibit conditional recruitment in the studied cases, but their preregistered target/rival score contributions have opposite signs. The warranted conclusion is therefore bounded: under explicit intervention semantics, FALI can distinguish mechanistic explanations that a common functional label alone would conflate.

Statements and Declarations

Competing Interests

Funding

Data Availability

Code Availability

Author Contributions

Use of Generative AI

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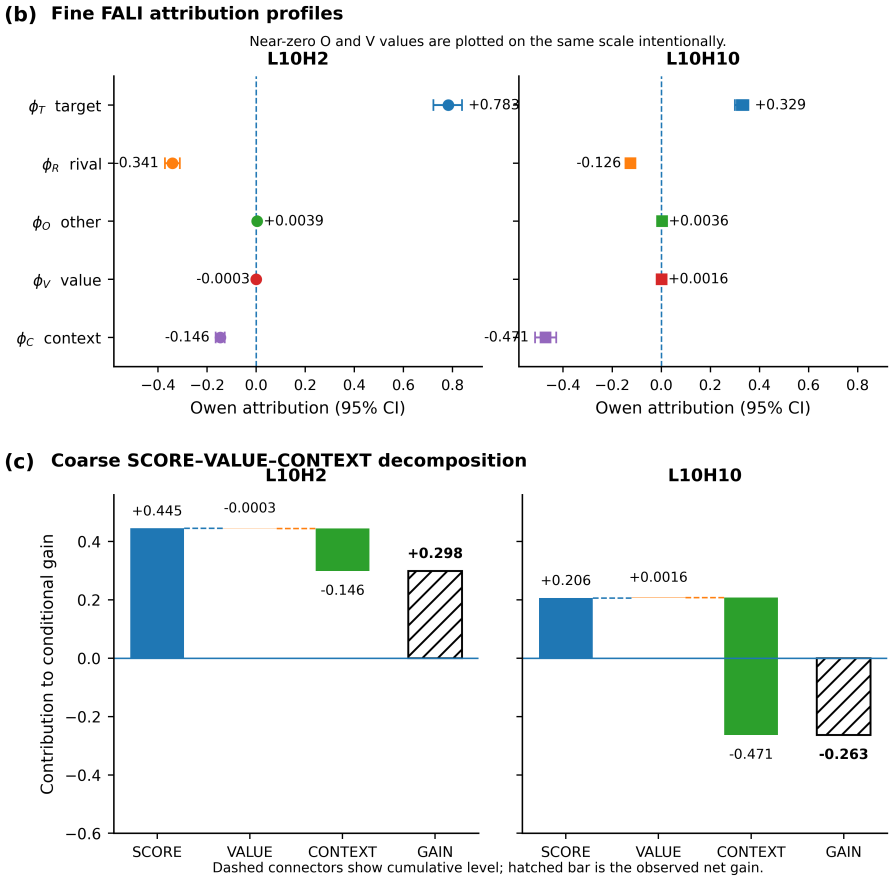


Fig. 4 Continued: Fine and coarse FALI decomposition in GPT-2 IOI. (b) Both L10H2 and L10H10 have positive target-score and negative rival-score contributions under the canonical FALI game; O and V remain near zero on the same plotted attribution scale. (c) The coarse SCORE/VALUE/CONTEXT decomposition explains why the same fine-score sign pattern does not imply the same functional response: L10H10 retains a positive SCORE contribution, but a substantially negative surrounding-context contribution outweighs it and produces a negative conditional gain. Points are exact balanced means; intervals in panel (b) are hierarchical 95% bootstrap confidence intervals. The context factor C denotes surrounding-network/context contribution under the specified intervention game and is not a uniquely localized downstream mechanism.

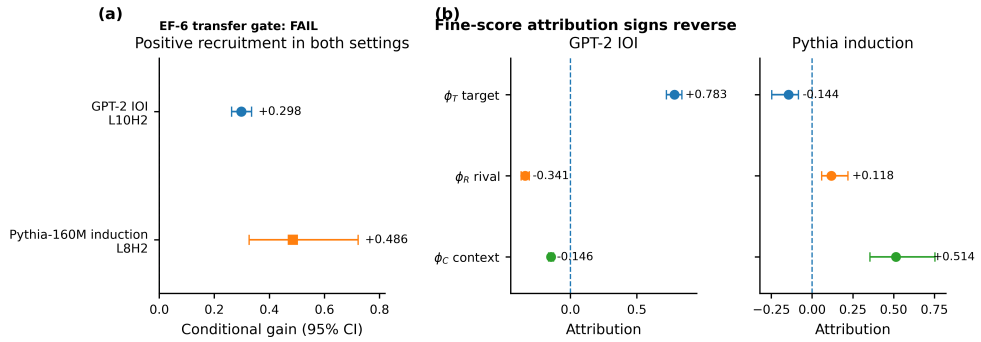


Fig. 5 Conditional recruitment occurs in both studied settings, while the preregistered fine-score mechanism does not transfer. (a) GPT-2 IOI L10H2 and Pythia-160M induction L8H2 both have positive conditional gain after their respective primary interventions. The numerical values are shown in their task-specific utility units and should not be interpreted as directly comparable effect magnitudes across tasks. (b) The observed fine attribution signs reverse: GPT-2 is target-positive and rival-negative, whereas Pythia is target-negative and rival-positive; their surrounding-context contributions also differ in sign. Points are exact balanced means and intervals are the preregistered 95% bootstrap confidence intervals for each experiment. The Pythia transfer test confirms induction behavior, primary-head importance, and positive conditional recruitment (G0–G2), but the predicted target-positive/rival-negative gates G3–G4 are not confirmed, yielding overall EF-6 Fail. This is a failed mechanism-transfer hypothesis, not a failed recruitment result.